

Model Run 1 - Tues 29 Jul 2025 (CNN)

You've successfully trained a Convolutional Neural Network (CNN) to identify different drum sounds (like kick, snare, hi-hat, etc.) within short audio segments. The model was trained on a large dataset of over 300,000 audio segments.

Here's what the key metrics from your test set evaluation mean:

- **Test Loss: 0.1170**
 - **In plain English:** This is a measure of how "wrong" your model's predictions are. A lower loss value means the model is making more accurate predictions. Your loss of 0.1170 is quite low, indicating that the model is performing very well in minimizing errors.
 - **For drum classification:** It means the model's overall predictions for whether a segment contains a kick, snare, hi-hat, etc., are very close to the actual labels.
- **Test Accuracy (Binary): 0.9531**
 - **In plain English:** This metric tells you the percentage of all individual drum sound predictions (e.g., "is it a kick?" "is it a snare?") that were correct. For example, if a segment has a kick and a hi-hat, the model needs to correctly identify both.
 - **For drum classification:** A binary accuracy of 0.9531 means that 95.31% of the time, the model correctly predicted whether a specific drum type was present or absent in an audio segment. This is a very high accuracy, suggesting the model is generally excellent at distinguishing between drum sounds.
- **Test Precision: 0.8886**
 - **In plain English:** Precision answers the question: "Of all the drum sounds the model *predicted* were there, how many were actually correct?" High precision means when your model says a drum sound is present, it's very likely to be correct, with few "false positives."
 - **For drum classification:** A precision of 0.8886 means that when your model identifies, say, a "snare drum," there's an 88.86% chance that it actually *is* a snare drum (and not another drum sound mistakenly identified as a snare). This is good for avoiding incorrectly transcribed notes in your sheet music.
- **Test Recall: 0.8261**
 - **In plain English:** Recall answers the question: "Of all the drum sounds that *were actually present*, how many did the model correctly identify?" High recall means the model is good at finding all the relevant drum sounds, with few "false negatives" (missed detections).
 - **For drum classification:** A recall of 0.8261 means that your model successfully identified 82.61% of all the actual drum sounds present in the audio segments. This is important to ensure that most drum hits are captured and appear in the generated sheet music.
- **Test AUC (Area Under the Receiver Operating Characteristic Curve): 0.9841**
 - **In plain English:** AUC is a comprehensive measure of a model's ability to distinguish between classes. A value of 1.0 means perfect classification, while 0.5 means random guessing. It's particularly useful for multi-label problems like yours.
 - **For drum classification:** An AUC of 0.9841 is exceptionally high. It indicates that your model has a fantastic ability to differentiate between the presence and absence of various drum sounds. It's very good at ranking which drum sounds are most likely present in a segment, even when there are multiple sounds occurring at once.

In summary: Your model is performing very well! The high accuracy and AUC, combined with strong precision and recall, suggest that your DrumScript's classification component is highly effective at identifying drum sounds, which is crucial for generating accurate musical sheet music.

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